Pulse splitting during self-focusing in normally dispersive media

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The self-focusing of femtosecond optical pulses in a normally dispersive medium is studied numerically. This situation represents a general problem that may be modeled by a 3 + 1-dimensional nonlinear Schrödinger equation, where two dimensions are self-focusing and the third is self-defocusing. The numerical simulations show that the dispersion causes the splitting of a pulse before it self-focuses into two temporally separated pulses, which then continue to self-focus and compress rapidly. The calculated behavior results in periodic modulation of the generated continuum spectrum, as was recently observed in continuum generation by focused femtosecond pulses in gases.

Self-focusing of optical pulses has been a topic of active research for some 20 years.[1–14] Stationary (cw) self-focusing has been well understood,[1,2] and the quasi-cw moving-focus model[3,4] has had much success in treating the self-focusing of nanosecond pulses. For self-focusing of shorter pulses, in a medium with a fast (electronic) nonlinearity, one expects that group-velocity dispersion (GVD) will become increasingly important. A primary consequence of self-focusing of short pulses is broad spectral generation, which has been attributed to self-phase modulation (SPM), four-wave mixing, and plasma production.[5] Recent observations of continuum generation in noble gases with weakly focused femtosecond pulses[6–10] have lead to the suggestion that self-focusing plays a critical role in this process as well. Periodic spectral modulation of observed single-shot continua from gases[6,8] was not attributed to SPM, as may have been the case for longer pulses and slower nonlinearities.[5] Instead, Strickland and Corkum[10] have suggested a model of a phased line array of continuum sources to explain this behavior. In addition, it was suggested in Ref. 10 that normal GVD results in the limitation of catastrophic self-focusing. In the past, numerical simulations performed to understand spectral broadening of short pulses during self-focusing have concentrated on plane-wave approximations and higher-order intensity effects.[11,12] The situation of self-focusing with an instantaneous cubic nonlinearity and normal GVD represents a general problem that, to lowest order, may be modeled by a 3 + 1-dimensional nonlinear Schrödinger equation, where two dimensions are self-focusing and the third (temporal) dimension is self-defocusing. The analysis from such an approach should then indicate the significance of GVD in self-focusing and the necessity of the inclusion of higher-order effects. This situation is an interesting contrast to the recently proposed total collapse of a self-focusing pulse for anomalous GVD,[13] i.e., a system with self-focusing in all three dimensions.

In this Letter a numerical integration of the spatiotemporal dependence of a self-focusing femtosecond pulse in a normally dispersive medium is performed in the slowly varying envelope approximation. The presence of GVD is found to cause the splitting of the original input pulse into two temporally separated pulses. The original pulse center does not self-focus, as the moving-focus model would predict, and the dramatic pulse-compression effect associated with self-focusing[14] occurs in the temporally split pulses away from the pulse center. This effect leads to the generation of two continua separated in time, which results in periodic spectral modulation, as was recently observed.[6,8]

The spatiotemporal dependence of the slowly varying field $\mathscr{E}(z,x,y,t) = \mathrm{Re}\{E(z,x,y,t)\exp[-i(\omega_0 t - n_0 k_0 z)]\}$, propagating along z in a dispersive medium with cubic nonlinearity is, to lowest order, governed by the 3 + 1-dimensional nonlinear Schrödinger equation:

$$\frac{\partial E}{\partial z} + i\frac{k^{(2)}}{2}\frac{\partial^2 E}{\partial t^2} - \frac{i}{2k}\left(\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2}\right) = in_2 k_0 |E|^2 E, \tag{1}$$

where $k \equiv n_0 k_0 \equiv n_0(\omega_0/c)$, $k^{(2)} = \partial^2 k/\partial\omega^2|_{\omega_0}$ is positive for normal GVD, and a retarded time frame traveling at $v_g = \partial\omega/\partial k|_{\omega_0}$ is assumed. This equation assumes the slowly varying envelope and paraxial ray approximations. The cubic nonlinearity is assumed to be instantaneous, and the intensity dependence of the group velocity, the shock term[15] $(2n_2/c)\partial(|E|^2E)/\partial t$, has been neglected. Equation (1) is solved numerically by using the split-step method and could be improved to account for nonparaxial rays,[16] but no significant effect was found for the case considered here.

Parameters typical of recent experiments in continuum generation are used. It is assumed that $\lambda_0 = 600$ nm and the medium is 40 atm of Xe[$k^{(2)} = 5 \times 10^{-3}$ ps^2/m (Ref. 17) and $n_2 = 3.6 \times 10^{-17}$ cm^2/W (Ref. 8)]. The input pulse ($z = 0$) is taken to have a sech^2 temporal intensity profile with a FWHM of 50 fs, a transverse Gaussian profile of FWHM 500 μm, and an initial phase

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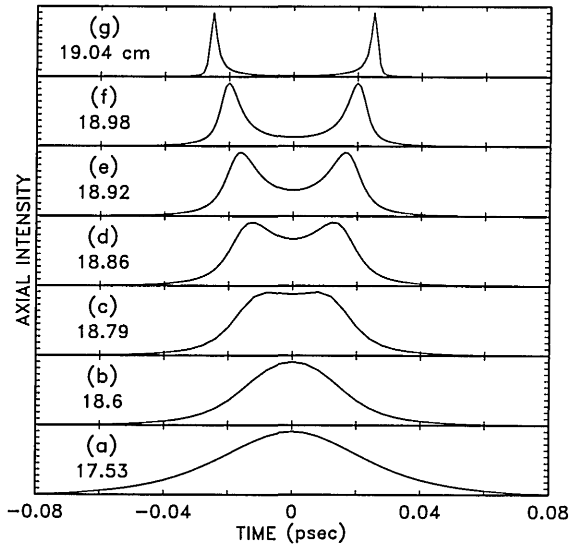

Fig. 1. Calculated intensity on axis versus time of a self-focusing pulse at the propagation distances indicated. The peak intensity ($\times 10^{12}$ W/cm^2) at each distance is (a) 0.85, (b) 6.3, (c) 11, (d) 14, (e) 20, (f) 34, and (g) 150.

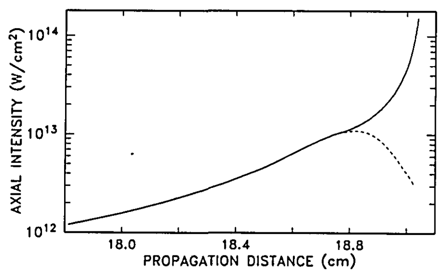

Fig. 2. Peak axial intensity of the split pulses (solid curve) and the axial intensity at the original pulse center ($t = 0$, dashed curve) versus the propagation distance.

curvature corresponding to a 25-cm focal length (thus the *f*-number is ~200). The peak input power of 45 MW was chosen to be three times larger than the critical power for the catastrophic collapse of a Gaussian beam,[2] $P_{\text{crit}} = (1.22)^2 \pi \lambda_0{}^2/32 n_0 n_2 =$ 15 MW, where the units of n_2 are in centimeters squared per watt. The calculation uses a three-dimensional grid with 192^2 transverse and 256 temporal points.

The calculation shows that the gentle focusing of the pulse results in approximately linear propagation initially, during which the pulse broadens to ~80 fs (at $z = 12$ cm) owing to GVD. The nonlinearity then begins to compress the pulse temporally through the self-focusing pulse-compression effect.[14] Figure 1 shows the pulse intensity on axis versus time and propagation distance in the vicinity of the self-focusing singularity. The surprising result is that in contrast to the expectations of the moving-focus model, the central part of the pulse does not self-focus earliest in the propagation. Rather the input pulse splits into two distinct temporal pulses, which then compress sharply to much higher intensity as they self-focus further. The quasi-cw moving-focus model[4] predicts that the peak of the pulse, having the largest power, should reach a singularity at the shortest self-focusing length, and the regions of lower power around the peak should self-focus further downstream. Qualitatively similar behavior to that shown in Fig. 1 was obtained over the complete range of parameters investigated: peak powers of as much as 15 times P_{crit} and GVD four times smaller were used.

The focal properties are seen more clearly in Fig. 2, which displays the peak intensity of the split pulses on axis (solid curve) and the axial intensity at the original pulse center ($t = 0$, dashed curve) versus propagation distance. It is clear that the original pulse center goes through a much weaker focus than the split pulses do, for which this calculation cannot follow further. The focus of the pulse center reaches a maximum intensity of $\sim 10^{13}$ W/cm^2, which corresponds to a focal FWHM of ~11 μm (19λ). At the highest intensity shown ($\sim 1.5 \times 10^{14}$ W/cm^2), the split pulses have a FWHM of 3 μm (5λ).

The physical explanation of this phenomenon is illustrated in Fig. 3, which shows the spatially integrated (over x and y) power as a function of time and propagation distance (solid curves) and, for comparison, the axial intensity from Fig. 1 (dashed curves). As the pulse nears the self-focusing distance and the nonlinearity increases, the region near $t = 0$, having the largest intensity, undergoes the most SPM. Because of normal GVD this self-phase-modulated energy near $t = 0$ broadens temporally, but this is not yet apparent in Figs. 1(a) and 1(b) because the strong self-focusing pulse-compression effect[14] dominates and continues to compress the axial intensity temporally near $t = 0$. However, energy does flow away from the peak and thereby reduces the integrated power near $t = 0$, as is apparent in the dip formed in the integrated power in Fig. 3(b). Thus self-focusing causes off-axis energy to flow toward the axis most rapidly at $t = 0$, but then GVD causes this energy to spread away from $t = 0$, reducing the integrated power. This continues until the inte-

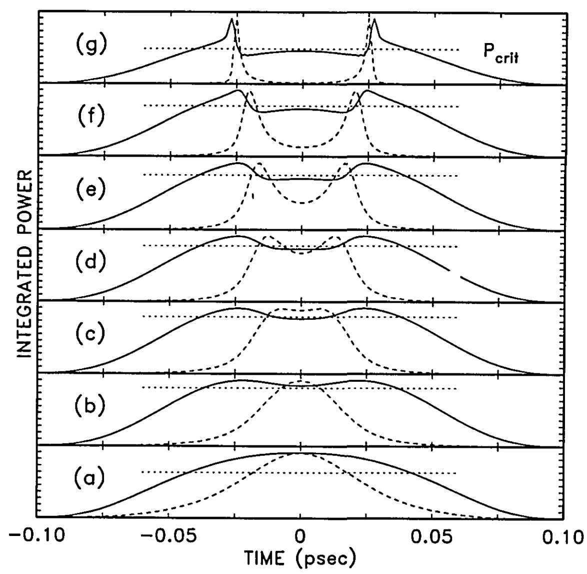

Fig. 3. Spatially integrated power (solid curves) at the same distances as in Fig. 1. The dashed curves are the axial intensities displayed in Fig. 1. The dotted lines indicate the critical power level (15 MW).

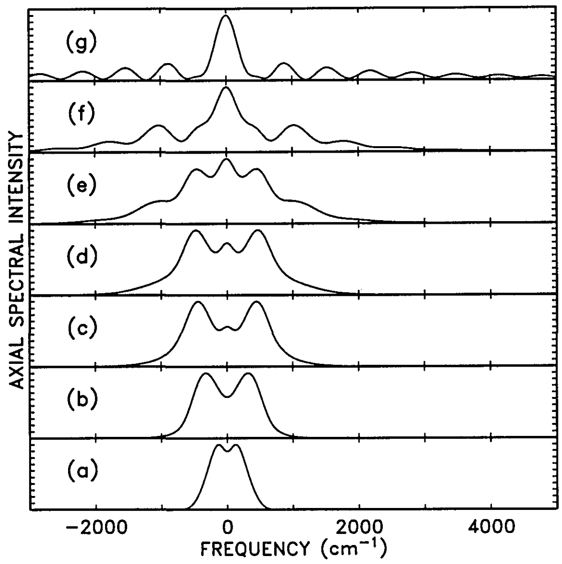

Fig. 4. Spectral intensity versus the relative frequency at the same distances as in Fig. 1.

grated power is reduced below P_{crit} (indicated by the dotted lines in Fig. 3), thereby terminating the self-focusing collapse of the pulse center [Fig. 3(c)]. Once past the focus for $t = 0$, the off-axis energy attempts to focus the temporal regions at $t \neq 0$, but each split pulse undergoes strong SPM and, through GVD, pushes the power further out from $t = 0$, leaving in their wake a broadening region with an integrated power less than P_{crit} [Figs. 3(d)–3(g)]. In Fig. 3(g) the integrated power is piling up near the split pulses, which appear to be forming an asymmetric shock. An interesting open theoretical question is whether the each split pulse collapses to a singularity or splits again owing to GVD.

As seen in Fig. 1, the intensity spikes have compressed to a pulse width of a few femtoseconds, which violates the slowly varying envelope approximation invoked initially. In addition, in this regime the shock term[15] is also undoubtedly significant and could lead to asymmetry in the generated spectra.[12] It is also observed[7,8] that saturation of the nonlinearity is significant for intensities in the range of 10^{13}–10^{14} W/cm^2. Whatever the limiting mechanism, one expects that the two intensity spikes would generate two similar continua separated in time, which then interfere to form spectral modulation. This behavior is seen in Fig. 4, which displays the spectral intensity on axis as a function of the propagation position. Initially one sees the development of a typical SPM spectrum. As the pulse splits and compresses the spectrum broadens dramatically and develops a spectral beat. The period of this modulation is then inversely related to the temporal separation of the intensity spikes. To find the spectral beat period at the cell output, one needs only the temporal separation of the two intensity spikes at the point in the propagation that the nonlinearity is extinguished by beam expansion. Since the nature of the limiting mechanism is unknown, quantitative comparison to experimental results is difficult. However, an estimate of this separation can be made based on the full width of the pulse (including GVD) over which the integrated power is larger than P_{crit}. For typical experimental conditions (focal distance ~30 cm and peak input power 2–3 times P_{crit}) this temporal width varies from approximately the pulse width (50 fs) at lower pressure (less than 10 atm of Xe) to ~150 fs for 40 atm of Xe, with the respective spectral beat periods being 670 and 220 cm^{-1}, which is in general agreement with measured spectral modulation.[8,10]

In conclusion, it has been shown that femtosecond pulses will split before the culmination of self-focusing (as opposed to the moving-focus model), which leads to the self-focusing of a double pulse. This is manifested by periodic spectral modulation owing to the generation of two temporally separated continua. The slowly varying envelope approximation may break down even in the presence of GVD, and further pulse splitting may occur, and it appears that higher-order effects need to be included.

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References

1. Y. R. Shen, Prog. Quantum Electron. **4,** 1 (1975).
2. J. H. Marburger, Prog. Quantum Electron. **4,** 35 (1975).
3. V. N. Lugovi and A. M. Prokhorov, Zh. Eksp. Teor. Fiz. **7,** 153 (1968) [JETP Lett. **7,** 117 (1968)].
4. M. M. T. Loy and Y. R. Shen, Phys. Rev. Lett. **22,** 994 (1969).
5. F. Shimizu, Phys. Rev. Lett. **19,** 1097 (1967); A. C. Cheung, D. M. Rank, R. Y. Chiao, and C. H. Townes, Phys. Rev. Lett. **20,** 786 (1968); R. R. Alfano and S. L. Shapiro, Phys. Rev. Lett. **24,** 592 (1970); A. Penzkofer, A. Laubereau, and W. Kaiser, Phys. Rev. Lett. **31,** 863 (1973); N. Bloembergen, Opt. Commun. **8,** 285 (1973).
6. J. H. Glownia, G. Arjavalingam, P. P. Sorokin, and J. E. Rothenberg, Opt. Lett. **11,** 79 (1986); J. H. Glownia, J. Misewich, and P. P. Sorokin, J. Opt. Soc. Am. B **3,** 1573 (1986).
7. P. B. Corkum, C. Rolland, and T. Srinivasan-Rao, Phys. Rev. Lett. **57,** 2268 (1986).
8. P. B. Corkum and C. Rolland, IEEE J. Quantum Electron. **25,** 2634 (1989).
9. T. R. Gosnell, A. J. Taylor, and D. P. Greene, Opt. Lett. **15,** 130 (1990).
10. D. Strickland and P. B. Corkum, Proc. Soc. Photo-Opt. Instrum. Eng. **1413,** 54 (1991).
11. T. K. Gustafson, J. P. Taran, H. A. Haus, J. R. Lifsitz, and P. L. Kelley, Phys. Rev. **177,** 306 (1969); F. Shimizu, IBM J. Res. Dev. **17,** 286 (1973); J. T. Manassah, P. L. Baldeck, and R. R. Alfano, Opt. Lett. **13,** 1090 (1988).
12. G. Yang and Y. R. Shen, Opt. Lett. **9,** 510 (1984).
13. Y. Silberberg, Opt. Lett. **15,** 1282 (1990).
14. J. H. Marburger and W. G. Wagner, IEEE J. Quantum Electron. **QE-3,** 415 (1967); G. L. McAllister, J. H. Marburger, and L. G. DeShazer, Phys. Rev. Lett. **21,** 1648 (1968).
15. F. DeMartini, C. H. Townes, T. K. Gustafson, and P. L. Kelley, Phys. Rev. **164,** 312 (1967).
16. M. D. Feit and J. A. Fleck, J. Opt. Soc. Am. B **5,** 633 (1988).
17. A. Dalgarno and A. E. Kingston, Proc. R. Soc. London Ser. A **259,** 424 (1961).